

Employee attendance and work hour monitoring are essential components of workforce management. However, most traditional systems currently in use—such as manual registers, RFID-based cards, or fingerprint biometric systems—are either outdated, error-prone, or inconvenient in practical usage, especially in large-scale or outdoor environments such as construction sites or warehouses.

These systems suffer from multiple limitations:

- **Manual systems** are prone to inaccuracies and often result in false or missed entries.
- **RFID cards** require employees to actively swipe the card, which may be forgotten or misused (e.g., buddy punching).
- **Biometric systems** are maintenance-heavy, require physical contact, and may be unsanitary or non-functional in dusty or humid environments.

Moreover, most of these systems lack remote accessibility, real-time data synchronization, and automatic work-hour calculation. Data often needs to be manually exported or processed before it can be analysed or used for payroll.

The goal of this project is to develop a **BLE-Based Employee Tracking System** that overcomes these challenges by offering:

- **Automatic detection** of employee entry/exit using BLE beacons without manual check-in or touchpoints.
- **Real-time logging** and syncing of attendance data via Wi-Fi to a cloud backend (Firebase).
- **Calculation of total active work hours** for each employee on a daily basis.
- **Remote accessibility** through a responsive web dashboard for monitoring from any location.

This system will help minimize errors, reduce the administrative burden, increase transparency in attendance records, and improve overall productivity. It will be especially useful for small-to-medium businesses, construction sites, workshops, and offices where managing accurate employee records is often difficult due to the lack of reliable, low-cost automation.